MESOPOTAMIA · A MODERN AUDIBLE REALIZATION

# Sammû · nīš tuḫrim

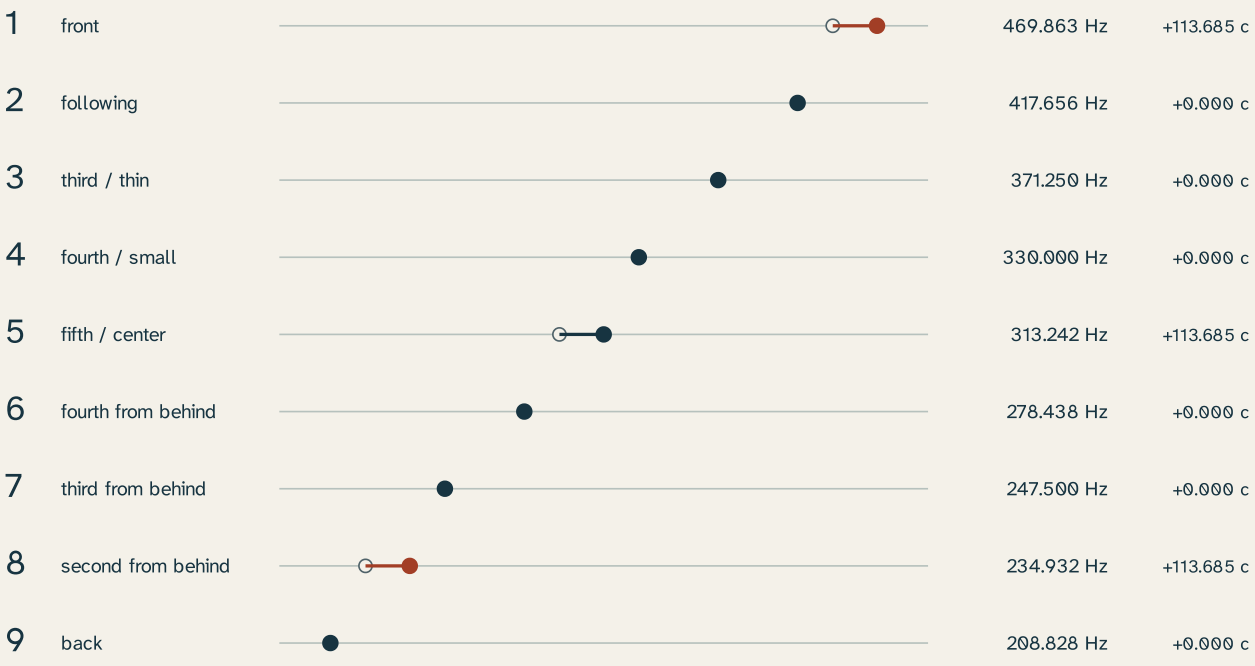
2 retuning moves · initial string 1: 440 Hz · companion strings: octave · sound: plucked

An audible fourths-and-fifths realization, not a recovered ancient performance. Six designated pairs are set to frequency ratios 4/3 or 3/2. Reference pitch, numerical precision, sound colour and the companion-string octave/unison choice are modern settings. The tablets do not record frequencies in hertz or these fractions.

## Sammû · nīš tuḫrim

Initial string 1: 440 Hz · moves: 2 · companions: octave

Open: initial · filled: current · rust: last changed · c: cents relative to each initial string




Modern numerical realization, not an excavated instrument or an ancient absolute-pitch claim.

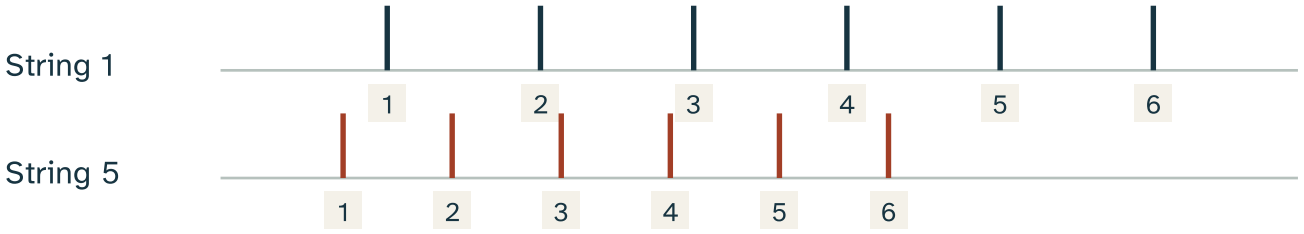
Six designated clear pairs use 4/3 or 3/2. Relative string 1: 2187/2048

## Hear the experiment

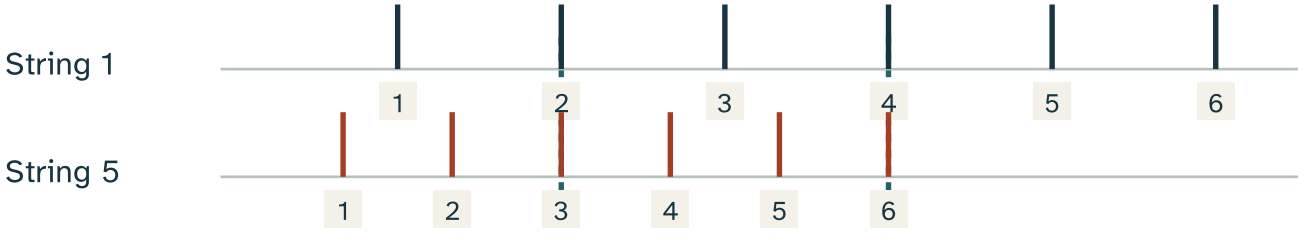
Before, then after: strings 1 + 5. Each sound is synthetic.

# The last change’s overtones

## Before · 0 shared partials




## After · 2 shared partials




0 Hz 3017 Hz →

Marks are the sine partials in this synthetic sound. A guide connects partials at exactly the same frequency; it does not establish what the ancient term “clear” meant.

# Current pitches

Exact ratios are relative to the initial first string, not a new reference after every move. Shifts compare each string with its own initial pitch.

| String | Ratio to initial 1 | Hertz | Shift ratio | Shift, cents |
|---|---|---|---|---|
| 1 | 2187/2048 | 469.863281 | 2187/2048 | 113.685006 |
| 2 | 243/256 | 417.656250 | 1/1 | 0.000000 |
| 3 | 27/32 | 371.250000 | 1/1 | 0.000000 |
| 4 | 3/4 | 330.000000 | 1/1 | 0.000000 |
| 5 | 729/1024 | 313.242188 | 2187/2048 | 113.685006 |

| String | Ratio to initial 1 | Hertz | Shift ratio | Shift, cents |
|---|---|---|---|---|
| 6 | 81/128 | 278.437500 | 1/1 | 0.000000 |
| 7 | 9/16 | 247.500000 | 1/1 | 0.000000 |
| 8 | 2187/4096 | 234.931641 | 2187/2048 | 113.685006 |
| 9 | 243/512 | 208.828125 | 1/1 | 0.000000 |

## Every retuning step

| Step | Action | Resulting tuning | Strings changed | String 1 ratio |
|---|---|---|---|---|
| 0 | initial | išartum | — | 1/1 |
| 1 | tighten | qablītum | 5 | 1/1 |
| 2 | tighten | nīš tuḫrim | 1 + 8 | 2187/2048 |

## How to read this model

Six pairs of strings are assigned exact 4/3 or 3/2 frequency ratios. The remaining diagnostic pair is left unclear. Each operation changes only the prescribed string (and its companion for strings 1 or 2), making that diagnostic pair clear. The next diagnostic pair then becomes unclear.

Seven moves in one direction return the tuning name but shift every string by 2048/2187 when loosening, or 2187/2048 when tightening. These fractions are consequences of this model, not ratios written on a tablet. No ancient reference pitch, recorded performance, instrument shape, or exact ancient sound colour is claimed.

The two tuning manuscripts from Ur are fragmentary. Combining them with later string-pair evidence is a scholarly reconstruction. In Mirelman's edition, supplied scribal omissions use ⟨ ⟩ and text restored across a break uses [ ]. For example, the first loosening case has an editorially supplied description of the diagnostic pair; the third is mostly restored.

## Sources and provenance


1. Sam Mirelman · Tuning Procedures in Ancient Iraq (2013)

   Analytical Approaches to World Music 2(2), 43–56. String names: p. 46; two manuscripts and uncertain instrument: pp. 47–48; preserved/restored clauses: pp. 49–52; seven tunings: fig. 5, p. 52; tightening: pp. 53–54.

2. Jay Rahn · Was Mesopotamian Tuning Diatonic? A Parsimonious Answer (2022)

   Music Theory Online 28.1. DOI 10.30535/mto.28.1.7. Paragraphs 9–16 distinguish physical strings from later pitch concepts and explain the reversible seven-name network; later sections examine assumptions behind an audible reconstruction.

3. Sam Mirelman & Theo J. H. Krispijn · The Old Babylonian tuning text UET VI/3 899 (2009)

   Iraq 71, 43–52. Edition of the second manuscript. Read with UET VII 74; the two witnesses are separate tablets, not joining fragments of one tablet.

4. CDLI P347039 · UET VII 74

   Catalogue record for one tuning manuscript from Ur, Old Babylonian period. The British Museum witness discussed by Mirelman is a cast; do not infer the original tablet's present location from the catalogue's collection field.

5. CDLI P254475 · CBS 10996

   Later witness for named string pairs. The catalogue checked on 22 October 2026 assigns Middle Babylonian, approximately 1400–1100 BCE; Mirelman (2013) used an older Neo-Babylonian dating. Combining witnesses is a scholarly reconstruction across time.

Reopen the experiment in Sammû

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